TEXAS TECH UNIVERSITY — SHARE LAB STEM NIGHT

Coupled Electro-Thermal EV Battery Physics

Mathematical Formulations, 4-Graph Dynamics Analysis & Software Architecture Reference Guide

1. Executive Summary & Lab Overview

In the SHARE Lab EV Rover Mission, students assume the role of an Electric Vehicle (EV) pit-crew engineer. Their objective is to select the exact charging current (**I_charge** in mA) required to charge a depleted Lithium-Ion battery pack from **20% to 100% State of Charge (SOC)** within a randomized time limit (20 to 40 minutes) without exceeding the thermal safety cutoff of **61.85°C (335.0 K)**.

The host console features a **Battery Dynamics & Graph Inspector** that visualizes the four synchronized state curves for every completed mission:



Figure 1: Host Console 4-Graph Inspector showing Run #2 (3950 mA over 40-minute window).

2. Comprehensive Analysis of the 4 Graphs

Graph 1: State of Charge (SOC %) vs. Time

Governing Principle: Coulomb Counting Integration.

Under constant-current charging, the rate of charge entering the cylindrical cell is strictly constant:

```
d(SOC)/dt = I_charge / Q_b ==> SOC(t) = SOC_0 + (I_charge / Q_b) * t
```

Because both charging current *I* and rated capacity *Q_b* (8,280 Amp-seconds / 2.3 Ah) are constant, the curve rises in a **purely linear slope** from initial 20% to 100%, where it levels off into a horizontal ceiling.

Graph 2: Coupled 3-Node Temperatures (T1 Core, T2 Surface, T_inf Coolant)

Governing Principle: Coupled 3-node thermal lumped masses.

Shows three temperatures:

- **T1 Core (°C):** Internal electrode jelly-roll temperature.
- **T2 Surface (°C):** Outer metal battery casing temperature.
- **T_inf Coolant (°C):** Temperature of the cooling jacket / fluid loop.

All three curves start at ambient (31.85°C / 305 K) and ascend in parallel, near-lockstep diagonal lines toward the 61.85°C safety limit.

Graph 3: Terminal Voltage (Vt) vs. Open-Circuit Voltage (Vocv)

Governing Principle: Discrete 10,001-point empirical chemical lookup plus series Ohmic resistance drop:

```
V_t(t) = V_ocv(SOC(t)) + I_charge * R_s
```

V_ocv comes from empirical testing (`SOC-OCV.xlsx`). Between 20% and 80% SOC, the chemical potential forms a flat plateau (~3.3 V to 3.45 V). Above 90% SOC, it enters a sharp chemical saturation "knee", shooting up to ~3.7 V - 4.1 V. Because

*I_charge * R_s* is constant, *V_t* runs parallel to *V_ocv*, shifted upward by ~0.20 V.

Graph 4: Electro-Thermal Heat Generation Rate (Q)

Governing Principle: Joule heating plus reversible entropic heating:

```
Q(t) = R_s * I_charge^2 - (dVocv / dT) * I_charge * T_1(t)
```

Joule heating (*R_s * I^2*) accounts for ~90% of heat (~1.2 W) and is constant because current is constant. The reversible entropic term is tiny (~0.1 W). Thus, total heat generation rate is a flat horizontal line throughout the charging run.

3. Mathematical Proof: Why Graph 2 Temperature Curves Are Linear

In classical heat transfer, temperature curves usually bend exponentially (*1 - e^(-t/tau)*). Why do these curves look like straight diagonal lines?

The Total Heat Capacity Integration Proof Summing the three thermal nodes of the pack:
C_total = C1 + C2 + C_inf = 51.9 + 6.2 + 10.0 = 68.1 J/K
Net heat input: **Q_net = Q_in - Q_out = Q - Q_c = 1.20 W - 0.02 W = 1.18 W (CONSTANT)**

Because the model has no ambient convective dissipation (no -h*A*(T - T_room)), net heat input is constant:
dT/dt = Q_net / C_total = 1.18 W / 68.1 J/K ≈ 0.0173 K/s ≈ 1.04 °C/min (CONSTANT)

Integrating a constant rate yields a **strictly linear ramp**:
T(t) = T_0 + (1.04 °C/min) * t

Furthermore, the core-to-surface thermal time constant is fast (**tau_1 = R1 * C1 = 2.64 * 51.9 ≈ 137 seconds / 2.2 minutes**). Within the first 2 minutes, a steady delta of *T1 - T2 ≈ Q * R1 ≈ 3°C* forms, after which all three curves rise in parallel lockstep.

Over a 30 to 40 minute chart, that 2-minute transient occupies only ~5% of the width, making the curve appear completely linear.

4. Mathematical Parameters & Governing Differential Equations

Parameter	Symbol	Numerical Value	Units	Description
Nominal Battery Capacity	Q_b	8,280.0	A·s (Coulombs)	2.3 Ah (2,300 mAh) Lithium-Ion cylindrical cell
Ohmic Series Resistance	R_s	0.050	Ohms (Ω)	Internal battery series resistance
Core Thermal Resistance	R_1	2.64	K/W	Core-to-surface conduction resistance
Surface Thermal Resistance	R_2	3.46	K/W	Surface-to-coolant conduction resistance
Core Heat Capacity	C_1	51.9	J/K	Thermal mass of internal jelly-roll
Surface Heat Capacity	C_2	6.2	J/K	Thermal mass of outer metallic casing
Coolant Heat Capacity	C_inf	10.0	J/K	Thermal mass of coolant jacket reservoir

Entropic Coefficient	dVocv/dT	1.0 × 10^-4	V/K	Reversible electrochemical temperature derivative
Coolant Heat Extraction	Q_c	0.020	Watts (W)	Active coolant loop heat extraction constant
Initial Ambient Temp	T_0	305.0	Kelvin (K)	Ambient initial temperature (31.85°C)
Max Safe Temperature	T_max	335.0	Kelvin (K)	Overheat cutoff threshold (61.85°C)

Discrete Integration Equations (Ts = 0.25 s)

```
SOC[k+1] = clip( SOC[k] + Ts * (I_charge / Q_b), 0.0, 1.0 )
V_ocv[k] = OCV_TABLE[ floor(SOC[k] * 10000) ]
V_t[k] = V_ocv[k] + I_charge * R_s
Q[k] = R_s_eff * I_charge^2 - (dVocv/dT) * I_charge * T1[k]
T1[k+1] = T1[k] + Ts * ( -(T1[k] - T2[k]) / (R1 * C1) + Q[k] / C1 )
T2[k+1] = T2[k] + Ts * ( -(T2[k] - T1[k]) / (R1 * C2) - (T2[k] -
Tinf[k]) / (R2 * C2) )
Tinf[k+1] = Tinf[k] + Ts * ( -(Tinf[k] - T2[k]) / (R2 * Cinf) - Q_c
/ Cinf )
```

5. Software & Coding Implementation

- **server/battery_physics.py:** Implements pure-Python parsing of `SOC-OCV.xlsx` (using standard library zipfile and XML elements without pandas

dependency). Runs forward Euler integration at 0.25s time steps and sub-samples the trajectory to exactly 101 points mapped to the 10-second web animation.

- **server/server.py:** Exposes `/api/battery/simulate` and real-time Server-Sent Events (SSE) on `/api/events`. Automatically persists the 60 most recent full trajectories to `server/game_history.json`.
- **webapp/src/components/MasterHostConsole.tsx:** Renders the 4 graphs onto hardware-accelerated HTML5 canvases using Retina High-DPI scaling (`devicePixelRatio`). Features dynamic Y-axis bounds, unit labels, dashed reference ceilings, and automatic Dark/Light theme switching.

6. STEM Night Presenter Guide (Key Talking Points)

1. **The Double-Edged Sword of Current:** More current charges the car faster (Graph 1), but heat grows as Current SQUARED (Graph 4: $Q \propto I^2$).
2. **The Insulated Bucket Analogy:** The temperature rises steadily (Graph 2) because we are pumping heat in faster than the small cooling pump can remove it—just like water filling a bucket at a constant rate!
3. **The Chemical Voltage Knee:** When Lithium-Ion batteries get near 100% full, ions crowd together, causing resistance and voltage to shoot up sharply (Graph 3).
4. **The Goldilocks Sweet Spot:** A successful engineer finds the current that hits 100% before time runs out while keeping the red core temperature line below the 61.85°C danger line!